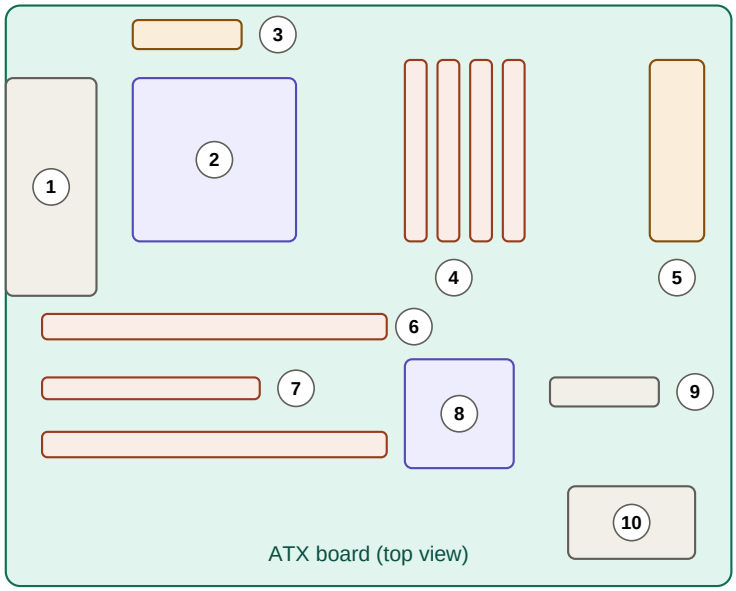
Motherboard Basics: Tasks and Answers

Task 1: Labeled ATX motherboard diagram

Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



Key

1. Rear I/O ports
2. CPU socket
3. 8-pin CPU power
4. RAM slots (DIMM)
5. 24-pin ATX power
6. PCIe x16 slots
7. PCIe x1 slot
8. Chipset
9. USB header
10. SATA ports

Task 2: Three popular CPU socket types

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

Socket	Maker	Compatible CPU example
LGA1151	Intel	Core i7-8700K (8th gen)
AM4	AMD	Ryzen 5 5600X
LGA1200	Intel	Core i5-10400F (10th gen)

Note: LGA1151 has two versions. 6th/7th-gen CPUs and 8th/9th-gen CPUs are not interchangeable, so always check the board's CPU support list.

Task 3: CPU and chipset, and how they work together with Spotify

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

CPU: It executes program instructions and does the calculations.

Chipset: It is the motherboard's traffic manager. It connects the CPU to storage, USB ports, network, audio, and expansion slots, and controls how data moves between them.

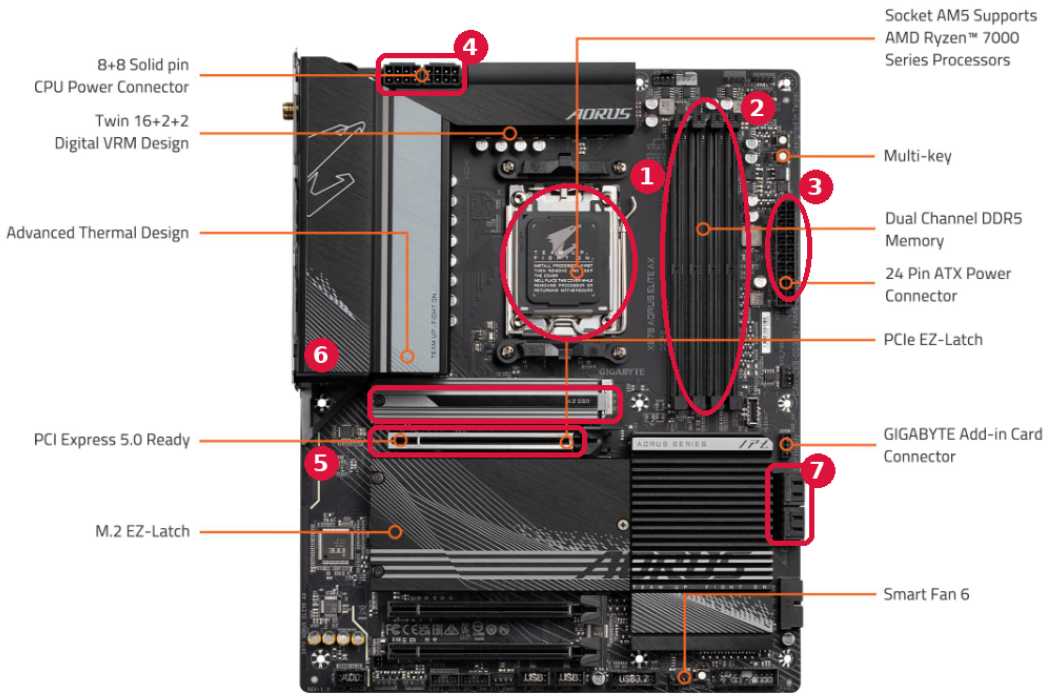
When you open Spotify:

1. The CPU runs the app's code and handles your click.
2. The chipset routes the request to the SSD or HDD to load the app files, and to the network chip (Wi-Fi or LAN) to reach Spotify's servers.
3. The streamed audio arrives via the network chip, and the chipset passes it to the CPU.
4. The CPU decodes the audio and the chipset sends it to the audio chip, which plays it through your speakers or headphones.

Task 4: Real motherboard photo with labeled ports

Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

Photo: Gigabyte X670 AORUS Elite AX (socket AM5), with 7 parts circled and numbered.



Key

1. CPU socket (AM5)
2. RAM slots (DDR5 DIMM)
3. 24-pin ATX power
4. 8+8 pin CPU power
5. PCIe x16 slot
6. M.2 SSD slot
7. SATA ports